

STAGE-09 — FUNCTIONS IN C

Chapter 1: What is a Function?

SLIDE 1 — WHAT IS A FUNCTION?

Definition

A function is a named block of code used to perform a specific task.

Functions help us divide a large program into smaller and easier parts.

Simple Idea

Task → Function → Result

Analogy

Imagine ordering tea in a college canteen:

You order tea → Function call Tea master prepares tea → Function execution You receive tea → Result

Small Example

None

```
void greet() {  
    printf("Hello Student");  
}
```

Remember

Function = A small worker that performs one task.

SLIDE 2 — WHY DO WE USE FUNCTIONS?

Main Purpose

Functions make a program easier to write, understand, and manage.

Benefits

1. Reusability — Write code once and use it many times.
2. Less Repetition — Avoid writing the same code again.
3. Easy Reading — Divide a large program into small parts.
4. Easy Debugging — Find errors in a smaller section.
5. Easy Maintenance — Change one function when needed.

Basic → Professional

Small Program → Functions → Modules → Large Application

Example

```
None
void greet() {
    printf("Hello Student");
}
```

The `greet()` function can be called whenever the message is needed.

Remember

Write once → Call many times.

SLIDE 3 — FUNCTION AND GREEDY ALGORITHM

Are They the Same?

No. A function and a greedy algorithm are different concepts.

Function

A function is a block of code that performs a particular task.

Greedy Algorithm

A greedy algorithm is a problem-solving method that chooses the best available option at each step.

Simple Difference

Function → Performs the work Greedy Algorithm → Chooses the option

Can They Work Together?

Yes. A function can contain a greedy algorithm.

Example

```
None
void chooseBest() {
    printf("Choose the best option");
}
```

Analogy

- Function = Student doing the task.
- Greedy Algorithm = Student choosing the easiest task first.

Remember

Function is the worker. Greedy is the decision method.

SLIDE 4 — HOW DOES A FUNCTION WORK?

Three Main Steps

1. Define — Write the function and its statements.
2. Call — Use the function name.
3. Execute — The statements inside the function run.

Flow

```
None
START
↓
DEFINE FUNCTION
↓
CALL FUNCTION
↓
```

EXECUTE FUNCTION



TASK COMPLETED



END

C Example

None

```
#include <stdio.h>
```

```
void greet() {  
    printf("Hello Student");  
}
```

```
int main() {  
    greet();  
    return 0;  
}
```

Output

None

```
Hello Student
```

Important

A function executes when it is called.

Memory Flow

Define → Call → Execute → Result

SLIDE 5 — RECAP: FUNCTIONS IN C

What Did We Learn?

1. A function is a named block of code.
2. It performs a specific task.
3. Functions divide large programs into smaller parts.
4. They reduce repeated code.
5. They improve readability and debugging.
6. A function is different from a greedy algorithm.
7. A function can contain a greedy algorithm.
8. A function runs when it is called.

Final Example

None

```
void greet() {  
    printf("Hello");  
}  
  
int main() {  
    greet();  
    return 0;  
}
```

Funny Memory Trick

Think of a college group project:

- Function → Team member
- Function Call → Giving a task
- Function Execution → Doing the task
- Output → Completed work
- Greedy Algorithm → Choosing the easiest task first

One-Line Recap

A function is a reusable block of code that performs a specific task when called.

Functions in C — 5 Questions

1. Theory MCQ

Which keyword is used when a function does not return any value?

A. `int` B. `return` C. `void` D. `float`

💡 Hint: It means “no value”.

📖 Glossary: `void` — A keyword used when a function returns no value.

Answer: C. `void`

2. Output Guess

None

```
#include <stdio.h>

void greet() {
    printf("Hello");
}

int main() {
    greet();
    return 0;
}
```

What is the output?

💡 Hint: The function is called inside `main()`.

📖 Glossary: `Call` — Using a function to execute its code.

Answer:

None

`Hello`

3. Fill the Missing Code

None

```
#include <stdio.h>

----- greet() {
    printf("Welcome");
}
```

}

💡 Hint: Use the keyword for no return value.

📖 Glossary: Return — The value sent back by a function.

Answer: `void`

4. Jumbled Sentence

Arrange the words correctly:

function / performs / a specific task / A

💡 Hint: Start with the word “A”.

📖 Glossary: Specific — A particular or definite task.

Answer: A function performs a specific task.

5. Fill in the Blank

A function is a block of code used to perform a _____ task.

💡 Hint: It means a particular task.

📖 Glossary: Task — A piece of work that needs to be done.

Answer: `specific`